

(a) NLP encoding of 13 structured clinical variables

1 Tabular input

13 clinical variables

age	69
pack-years	48
ECOG	1
albumin	4.0
dNLR	2.4
histology	adeno
...	...

df_to_text

2 Sentence

English text prompt

"The patient is a 69-year-old,
48 pack-years, ECOG 1,
albumin 4.0, dNLR 2.4 ... "

3 Encoder

all-MiniLM-L6-v2

6 Transformer layers

self-attention

self-attention

self-attention

mean-pool · L2-norm

Multi-Head Attention

MatMul

Scale

SoftMax

MatMul

Linear

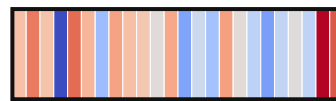
Q

K

V

4 Embedding

384-dim embedding

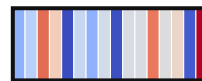


"NLP raw"

16-dim "NLP-PCA16"

PCA (16 comp.)

≈ 94% variance



5 Multimodal fusion

Fusion (DyAM)

+ CT radiomics
+ Pathology IHC
+ Genomics
+ PD-L1 TPS

attention-weighted risk

! no_scale=True — **RobustScaler disabled**
so the cosine-normalised geometry of the
embedding space is preserved.

embedding value

